Problem 9.19

A deposit is made on January 1, 2004. The investment earns interest at a rate equivalent to an annual rate of discount of 6%. Calculate the monthly effective interest rate for the month of December 2004.

Solution

Let X be the amount deposited on January 1, 2004, and let Y be the amount at the end of the year. Since the annual effective discount rate is $d = 0.06$, we have

$$X = Y(1 - 0.06).$$

Equivalently,

$$Y = X(1 - 0.06)^{-1}.$$

Let j be the monthly effective interest rate. Over twelve months,

$$Y = X(1 + j)^{12}.$$

Thus,

$$(1 + j)^{12} = (1 - 0.06)^{-1}.$$

Since $1 - 0.06 = 0.94$, we get

$$(1 + j)^{12} = (0.94)^{-1}.$$

Taking the twelfth root of both sides,

$$1 + j = (0.94)^{-1/12}.$$

Therefore,

$$j = (0.94)^{-1/12} - 1.$$

Finally,

$$j \approx 0.0051696.$$

Hence, the monthly effective interest rate for December 2004 is

$$\boxed{0.51696\%}.$$